

Beyond Words: Evaluating Verbal and Gestural Clarification Strategies During Ambiguous Robot Instructions

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Motivation:

People often provide ambiguous instructions to a robot. A robot can then act on its best interpretation, or request clarification. Prior research demonstrates that selective clarification, compared to always clarifying or never clarifying, can improve interaction efficiency [7], and that targeted follow-up questions can improve perceived robot competence [1]. Research on legible motion further shows that movement can help humans infer a robot's goal [2]. Recent VLA research integrates clarification dialogue with physical action [5].

Recent frameworks such as Ask-to-Clarify enable robots to resolve ambiguous instructions through multi-turn dialogue. However, its evaluation emphasizes task success, disambiguation accuracy, and inference latency, without directly assessing users' workload, frustration, or trust [5]. Although prior work explicitly considers reducing human assistance [6], these technical measures do not establish how users experience different clarification strategies. This motivates our comparison of verbal, gestural, and combined clarification, examining both interaction efficiency and users' perceptions of the robot.

We investigate whether a hybrid reach-pause-and-gesture can serve as an understandable, efficient confirmation request during a timed physical task compared to other forms of clarification or assumption. Our contribution is a controlled comparison of verbal confirmation, gestural confirmation, and hybrid (Ask-and-Gesture) confirmation, examining both task performance and human perceptions.

Research Questions and Research Hypotheses

Research Question: How do clarification strategies affect **task success rate**, **task completion time**, frustration, trust, and perceived competence?

Hypothesis: When the robot uses the designed gesture of approaching and confirming to clarify ambiguity, the users will notice the robot to be more efficient and human-like.

Proposed Methodology:

Simulation Development and Physical Deployment

We will first develop and test the system using Hello Robot's Stretch simulation stack in MuJoCo [3]. This stage will support implementation of task scenarios, the communication interface, clarification logic, arm gestures, and data logging.

After testing the four strategies in simulation, we will transfer the system to an available physical Stretch 2 or 3 robot. Physical testing will verify gesture visibility, motion execution, perception, and interaction timing before the main participant study. Simulation results will support system

development; conclusions about physical human–robot interaction will be based on the physical study.

Task-Based Interaction

Participants will complete a fixed set of tasks. Each participant will receive task descriptions or target images and communicate requests using their own wording.

User-to-robot communication will use either speech or a simplified chat interface. We will select one interface before the main study and keep it consistent across conditions. Participants will have a way to interrupt execution and indicate whether an episode succeeded or failed.

Completion time will be measured as an outcome.

Creating Ambiguity

We will arrange similar distractor objects, such as multiple cups and blocks, so that some requests may refer to more than one plausible target, similar to the experiment setup used in [1][7]. Participants will know their intended target, while the robot will receive only the information they communicate.

Participants will not be instructed or rewarded to provide ambiguous commands. A **pilot study** will assess whether the scenes and tasks produce sufficient ambiguity naturally, while avoiding clutter that primarily causes unrelated perception or grasping failures.

Interaction Conditions:

There are four conditions in which a robot will execute, after the robot receives an unclarified task instruction from the humans [Table 1].

1. Immediate action: Robot guesses the humans' intent and proceeds with the best hypothesized execution.
2. Asking humans for clarification: Robot asks a verbal question; user must communicate a response to the robot.
3. Arm gestures: Robot leans toward the hypothesized target object/location and pauses; user will nudge the robot to keep proceeding or make corrections to the robot's assumptions, before the robot takes an action.
4. Hybrid (Ask-and-Gesture) (2 + 3): Robot will simultaneously do the arm gestures while asking humans for clarification.

In 2, 3, and 4, a robot will wait until a response is received from humans.

Table 1: The Four Modes of Clarification

	No Gesture	Gesture
No Voice	Immediate Action	Arm Gesture
Voice	Asking for Clarification	Hybrid: Ask-and-Gesture

Proposed Evaluation:

Experimental Design: We use a **Factorial Design** structure.

- **Between-subjects Design:** Each participant will contribute data to only one of the four modes of clarification.
 - **Note:** If we are not able to recruit enough participants, we will conduct the experiment via **Within-subjects Design**. The robot randomly cycles through the four modes of clarification during the experiment, for **counterbalancing**.

Variables:

- **Independent Variables:** The four modes of clarification: Immediate action, asking humans for clarification, Arm Gestures, and a Hybrid Ask-and-Gesture.
- **Dependent Variables:** Objective (task completion time, task success rate, and number of clarification turns) and Subjective (user frustration, trust, and perceived intelligence).

Post-Results Survey: Participants will complete a survey assessing their experience with the robot. Open-ended questions will invite participants to explain what made the interaction easier or more difficult. If we use a within-subjects design, participants will provide strategy-specific ratings after each block. We will also assess whether participants interpreted the reach-and-pause as a request for confirmation. This manipulation check will be administered after the relevant interaction observations to avoid teaching participants the gesture’s meaning prematurely.

Analysis: Statistical analysis will match the final experimental design. Between-subjects comparisons will use independent-group methods, and a within-subjects study will use repeated-measures methods. Trial-level analyses will account for repeated observations from each participant, and success rates will be analyzed using methods appropriate for binary outcomes. We will report effect sizes and uncertainty estimates alongside significance tests, including findings that do not support the hypothesis.

Validity: Comparable task difficulty, object arrangements, robot capabilities, and response interfaces will support **internal validity**. Physical interaction and freely formulated requests support relevance to everyday robot use. However, the controlled task setting and participant sample limit **external validity**. We will not assume that results generalize to all household or workplace interactions.

Project Timeline:

- Project Check-In Day 1 (10/8) - Finalize VLA deployment on the simulation-based robot; build the gamified chat/audio GUI and scoring metric.
- Project Mid-Point “Poster” Session (10/20-22) - Have the 4 modes of clarification functioning; conduct a pilot test to verify that the urgency holds and the scoring metric correctly forces uncertainty.
- Project Check-In Day 2 (11/10) - Complete the experiments with all participants; administer the "Score Debrief" surveys.
- Final Project Presentations (12/1-3) - Finalize data analysis, concluding which mode of clarification yielded the best objective score and subjective user ratings.

Proposed Grading Rubric:

Finding / System Component / Capability / Validation Test	Points
	(__ /150)
Literature Review: Identifying the gap in VLA clarification costs and HRI deception methodologies.	(__ /15)
System Implementation: Building the experiment interface, potential scoring system, and the 4 physical expressions on Stretch/Simulation.	(__ /40)
User Study Execution: Successfully Designing and executing the study and maintaining the records for participants.	(__ /35)
Data Analysis: Statistically analyzing the mode of clarifications' success rates vs. survey.	(__ /35)
Final Report: Clarity, proper use of HRI terms, and presentation of findings.	(__ /25)

References

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- [7] David Whitney, Eric Rosen, James MacGlashan, Lawson L. S. Wong, and Stefanie Tellex. 2017. Reducing Errors in Object-Fetching Interactions through Social Feedback. In *2017 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE. [doi:10.1109/ICRA.2017.7989121](https://doi.org/10.1109/ICRA.2017.7989121).

Required Glossary Terms Used (8)

1. Within-subjects Design
2. Between-subjects Design
3. Mixed Methods Design
4. Factorial Design
5. Independent Variables
6. Dependent Variables

7. Likert scale
8. External validity
9. Internal validity